Arista Access Node ELEC-ACC-01 Standard Operating Procedure (SOP) for Route Churn and Adjacency Loss

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9146

Description

This SOP outlines the procedures to be followed by the Telecom Network Architect (Senior MTS) when encountering a Major Severity alarm (ALARM_9146) on the Arista Access Node ELEC-ACC-01 at the Electronic City Hub. The failure domain of interest is the Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding). This failure domain is a critical concern due to its potential to cause widespread network instability and downtime. The root cause of this failure domain is often attributed to misconfiguration, hardware issues, or software bugs within the routing protocols. Specifically, BGP route churn may occur when there are frequent changes to the routing table, OSPF adjacency loss may result from issues with neighbor relationships, and ISIS LSP flooding may be caused by errors in the routing information database. To prevent such failures, it is essential to maintain a robust and well-configured network topology, implement regular monitoring and maintenance procedures, and ensure timely troubleshooting and resolution of any issues that may arise.

Pre-requisite Checks

- Check: Verify that the Electronic City Hub is accessible and there are no issues with the physical infrastructure.

- Check: Confirm that the redundant links are operational and functioning correctly.

- Check: Ensure that the remote hands are dispatched to the Electronic City Hub for assistance if required.

Potential Root Causes

Cause	Probability
BGP Misconfiguration	High
OSPF Neighbor Issues	Medium
ISIS LSP Errors	Low
Hardware Failure	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp all, Expected Output: BGP table version is 1, local router ID is 10.0.0.1, Description: This command displays the BGP table version, local router ID, and other relevant information to help diagnose BGP-related issues.
```

```
Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays the OSPF neighbor information, including the neighbor ID, priority, state, dead time, address, and interface.

Command: show isis database, Expected Output: IS-IS Level 2 Link-State Database, Description: This command displays the ISIS database information, including the level, LSP ID, sequence number, age, check value, and other relevant details.
```

Resolution Steps

Step 1: Step: Verify BGP Configuration, Command: show running-config | include bgp, Description: This command displays the BGP configuration, which can help identify any misconfigurations or issues that may be causing the route churn.

Step 2: Step: Reset OSPF Neighbor, Command: clear ip ospf process, Description: This command resets the OSPF process, which can help resolve issues with neighbor relationships and adjacency loss.

Step 3: Step: Flush ISIS LSP, Command: clear isis database, Description: This command flushes the ISIS LSP database, which can help resolve errors in the routing information database.

Post-Check Verification

```
Command: show ip bgp all, Expected Output: BGP table version is 1, local router ID is 10.0.0.1, Description: This command verifies that the BGP table version and local router ID are correct after the resolution steps.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command verifies that the OSPF neighbor information is correct after the resolution steps.

Command: show isis database, Expected Output: IS-IS Level 2 Link-State Database, Description: This command verifies that the ISIS database information is correct after the resolution steps.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: copy startup-config running-config, Description: This command restores the original configuration, which can help roll back any changes made during the resolution steps.